1 MODALITIES

No best modality: trade cost, capability, compatibility, comfort and radiation.

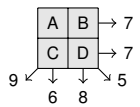
- **Ultrasound**: sound waves; **operator dependent**; no radiation, cheap; air and bone block the view. Generalises **worst** to new sites.
- **X-ray**: heated filament emits e^-, accelerated by high voltage onto a metal target. Bone bright; projection = structures superimposed.
- **CT**: X-ray tube + detector rotate, table moves → projections from many angles → 3D. Quick, anatomical, radiation. **Hounsfield units**: water 0, air −1000, bone +hundreds. Uses: **diagnosis and (surgery) planning**.
- **MRI**: B_0 aligns spins on z; 90° RF pulse tips them; **relaxation** back is the signal. Raw data = **k-space**. Best soft-tissue contrast, several weightings (T1, FLAIR); **slow, sedation in young children**; no radiation; most expensive; arbitrary intensity units.
- **PET / nuclear**: gamma rays, **physiological**, poor anatomy.

	rad.	inv.		rad.	inv.
US	N	N	PET	Y	min
X-ray	Y	N	contrast	maybe	min
CT	Y	N	endoscopy	N	Y
MRI	N	N	histopath.	N	Y

- **Doses**: chest X-ray 20 μSv; mammogram 0.4 mSv; head CT 2 mSv; chest CT 7 mSv; background ≈4 mSv/yr; **100 mSv/yr** lowest linked to cancer; 4 Sv fatal untreated.
- **Planes**: sagittal (left|right), coronal (front|back), axial (top|bottom).
- **Why AI**: more images; anomaly detection, triage, reconstruction; **quantification, not decision**: support clinicians.

Back-projection

Each pixel gets the sum of all rays through it. Value = (BP − total)/(#directions − 1).



BP: A = 7+6+5 = 18, B = 24, C = 22, D = 20; total = 7+7 = 14. **A=2, B=5, C=4, D=3** (check all 6 sums). Plain BP is blurry; more angles = sharper.

2 SEGMENTATION

Per-voxel classification with fully convolutional nets; preprocessing and configuration matter more than architecture.

Notation

$x \in \mathbb{R}^{M\times\Omega}$ (M modalities), classes $\mathcal{K} = \{0, \dots, K\}$.

$N(x;\theta) \in \mathbb{R}^{(K+1)\times\Omega} \xrightarrow{\text{softmax}} s_\theta \in (\Delta_K)^\Omega$

$\xrightarrow{\text{argmax}} S \in \mathcal{K}^\Omega \xrightarrow{\text{one-hot}} s \in \{0,1\}^{(K+1)\times\Omega}$.

- Δ_K: K+1 probabilities summing to 1; vertices = one-hot.
- Train on s_θ (argmax has no gradient); evaluate on one-hot s.
- S = argmax over the class axis, per pixel.

Architectures

- **CNN with FC head**: fixed input and output size.
- **FCN**: only conv / pool / up-conv / 1×1 → any input size. **No resize needed** with batch 1, no patching; size divisible by $2^{\#\text{pools}}$ (else pad).
- **U-Net** 572→388: unpadded 3×3 convs (−2 px each), *copy & crop* skips. "Same" padding → output = input size.
- **2D** cheap, no inter-slice context; **3D** (V-Net) heavy; **2.5D** (FastSurfer) in between.

Training data

- **Augmentation**: flip/rotation on image **and** label; noise on image only.
- **Offline**: $D_a = \bigcup_A \{A(D)\}$, $|D_a| = A \cdot N$ (storage). **Online**: new random draw each iteration.
- **Sub-patching**: $\mathcal{L} = \sum_b \mathcal{L}(\tilde{y}_b, \tilde{s}_b)$; net never sees whole image; patches may lack a class.

Post-processing

- **Connected components** (deck: "non-maxima suppression"): keep the largest. OK for one organ; **not** for multiple lesions.
- **DenseCRF**: energy = unary $-\log s_\theta$ + pairwise (close and similar-intensity pixels get the same label). Snaps to edges; weak with low contrast.
- **Ensembling**: better, but cost × number of models.

nnU-Net (self-configuring; loss fixed = CE + Dice)

- **Fingerprint** (after crop to non-zero): image size, **spacing**, **modality**, #classes, #cases, **foreground intensity** (mean, sd, 0.5/99.5 pct).
- **Normalisation**: CT → clip to fg 0.5/99.5 pct + *global* mean/sd. Others → per-image z-score $(x-\mu)/\sigma$.
- **Resampling**: per-axis **median** spacing (10th pct out-of-plane if max/min spacing > 3). Images: 3rd-order spline. Labels: one-hot → interpolate → argmax (interpolating class numbers directly invents classes: halfway 0 & 2 = 1).
- **Patch**: start at median image shape, shrink until **batch** ≥ 2 fits GPU memory. Cascade if patch < 12.5% of image.
- **Fixed**: plain U-Net, instance norm (batches tiny), leaky ReLU, deep supervision, SGD Nesterov 0.99, lr 0.01 poly, 1000 epochs, 1/3 patches forced foreground (imbalance), sliding window + mirror TTA.
- **Empirical** (5-fold CV): 2D / 3D / cascade / ensemble; keep largest CC?
- **Lesson**: configuration > architecture (KiTS top-15 all 3D U-Net).
- Config = deterministic rules on fingerprint + **GPU memory budget**: other GPU can give other patch, batch, depth.

3 LOSSES & GRADIENTS

CE = dense per-voxel signal, dominated by background; Dice = region overlap, robust to imbalance; combine them.

$\nabla_\theta = \frac{\partial \mathcal{L}}{\partial s_\theta} \cdot \frac{\partial s_\theta}{\partial l_L} \cdots \frac{\partial l_1}{\partial l_0}$, $\theta_{t+1} = \theta_t - \gamma \nabla_\theta$.

- **Usable loss**: differentiable w.r.t. the **network output**. Not w.r.t. label; need not be in [0, 1] or positive.

Cross-entropy

$\mathcal{L}_{CE} = \sum_i \sum_k -y^{(i,k)} \log s_\theta^{(i,k)}$, $\frac{\partial \mathcal{L}_{CE}}{\partial s^{(i,k)}} = -\frac{y^{(i,k)}}{s^{(i,k)}}$

- $-1/s$ if y=1, else 0: local, voxel-specific.
- → $-\infty$ **when** y=1, $s \to 0$ (confident mistake).
- $D_{KL}(P\|Q) = H(P,Q) - H(P)$ → minimising CE = minimising KL.
- **Von Neumann**: already used in statistical mechanics; and *"no one really knows what entropy really is, so in a debate you will always have the advantage"*.

Dice

$\text{DSC} = \frac{2|\Omega_y \cap \Omega_s|}{|\Omega_y| + |\Omega_s|} = \frac{2\sum ys}{\sum(y+s)}$

$\mathcal{L}_{DSC} = \frac{1}{|\mathcal{K}|}\sum_k \left(1 - \frac{2I}{U}\right)$, $I = \sum_i y\, s_\theta$, $U = \sum_i (y + s_\theta)$

$\frac{\partial \mathcal{L}_{DSC}}{\partial s^{(i,k)}} = \begin{cases} -2(U-I)/U^2 & y = 1 \\ 2I/U^2 & y = 0 \end{cases}$

CE vs Dice

- Dice gradient much **smaller** (divided by U^2).
- Dice gradient takes only **2 values** per class and image (global I, U); CE is voxel-specific.
- E.g. y = 1111 00, s = .2 .9 .8 .7 .6 .1: CE −5, −1.1, −1.25, −1.4, 0, 0; Dice −.18 (all fg), +.10 (all bg).
- **Combo** $\mathcal{L}_{CE} + \lambda\mathcal{L}_{DSC}$ (nnU-Net): dense signal + region signal. Families: distribution (CE, focal, TopK), region (Dice, IoU, Tversky), boundary, compound. Compound most robust; weighted CE worst; Tversky α=β=.5 = Dice; focal $(1-s)^\gamma$ down-weights easy voxels.
- CE value: $p_{\text{true}} = 0.8 \to -\ln 0.8 \approx 0.22$; $p = 0.5 \to 0.69$.

Dice NaN fix (empty class)

Empty class → 0/0 → NaN, and `.mean()` spreads it.

```
eps = 1e-10
return (1-(num+eps)/(denom+eps)).mean()
```

Weak supervision

- Labels: tags < dots < scribbles < boxes < masks.
- **Proposal-based** $\min_{\theta,y_\theta}$ (DeepCut): trains on its own masks → **shrinks to empty** (box only used at init).
- **Direct**: $\min \mathcal{L}(\tilde{y}, s_\theta) + \lambda R(s_\theta)$: partial CE on labelled pixels (unlabelled get no gradient) + CRF; **tightness prior**: object touches all 4 box sides → every row/column in box has fg.
- **Boundary loss** $\sum s\, \phi_{\partial\Omega_y}$: gradient = distance map; good for imbalance.

4 REGISTRATION

Align moving to fixed by minimising a dissimilarity; transform complexity follows anatomy and time.

Fixed F, moving M, $\min_T \mathcal{L}(F, T(M))$.

SSD same intensities: same modality (CT–CT)
NCC linear relation: brightness changes
NGF gradient directions: multimodal edges
MI statistical dependency: **multimodal** (CT–MRI)
MIND self-similarity: very different modalities

- **MI** = $\sum p(a,b) \log \frac{p(a,b)}{p(a)p(b)} = H(F) + H(M) - H(F,M)$ (joint entropy). Aligned → sharp joint histogram → high MI.
- **Affine matrix** (in header, image → world coords): **voxel size, position in the real world, rotation** (+shear). Not rigidity/elasticity. $A = C \cdot R \cdot S \cdot T$. Rigid 2D/3D = 3/6 params, affine 6/12. Frames: world, patient, image.
- **Elastic**: displacement per voxel; ill-posed → $\min \mathcal{L} + \lambda R(T)$ (smoothness, no folding). Steps: centre of mass → translation → affine → deformable; coarse-to-fine pyramid.

Which registration?

- **Bones** (adult, short interval): rigid (femur, skull); ribs either (cage moves with breathing).
- **Soft/moving organs**: non-rigid (lungs breathing, heart beating, kidneys).
- **Growth** (children/teens over years): **everything non-rigid**.

Learned registration (VoxelMorph)

- U-Net $g_\theta(F, M) = \phi$ → spatial transformer $M \circ \phi$ → loss $\mathcal{L}_{sim}(F, M\circ\phi) + \mathcal{L}_{smooth}(\phi)$.
- **No labels needed** (unsupervised). Optional: segmentation or landmark term $\mathcal{L}_{seg}(s_f, s_m \circ \phi)$.
- Learned vs classical: + **fast inference** (one pass); − **no guarantee of correctness**, depends on training data.
- **Seg U-Net** → **registration**: 2 input channels (F, M); 3 output channels, **no softmax**; add spatial transformer; loss sim + smooth.

5 METRICS

The metric follows the clinical question; combine overlap + boundary; conventions for corner cases bias results; annotations are approximations.

Choosing metrics (Metrics Reloaded)

- Failures: wrong problem category; bad metric choice (Dice on tiny objects); bad application (average images, not patients).
- Segmentation: overlap (Dice/IoU) + boundary (**NSD**). Per image, then aggregate. **Don't inherit the metric from the loss.**
- Detection has no TN → no accuracy, specificity, AUROC.

Dice properties

- **Favours over-segmentation**: same N errors, over $\frac{2|G|}{2|G|+N}$ > under $\frac{2|G|-2N}{2|G|-N}$ ($|G|$=100, N=10: .952 vs .947). Worse for small objects.
- Blind to **shape** and to **space** (shuffled pixels: Dice stays 0.92, HD changes).
- Jaccard $J = \frac{D}{2-D}$, $D = \frac{2J}{1+J}$, $J \le D$. Dice = pixel-level F1.

Dice corner cases

	GT not empty	GT empty
pred. not empty	formula	0
pred. empty	0	0/0 undefined

$\frac{2|\cap|+\varepsilon}{|y|+|s|+\varepsilon}$ → empty/empty = 1 → **bias: absent structures count as perfect, average inflated**.

Reporting

- Without post-processing to judge the method, with it for the deployed system; ideally both.
- Define a *sample* before testing (fold/scan; voxels = p-hacking).
- Inter-rater ≈ 90% DSC → small differences meaningless. Significant figures: 7 ± 0.5 → write 7, not 7.00.

Classification metrics

Sens $\frac{TP}{TP+FN}$, Spec $\frac{TN}{TN+FP}$, Prec $\frac{TP}{TP+FP}$, F1 $\frac{2PR}{P+R}$.

- **MCC** ∈ [−1, 1], robust to imbalance, multi-class.
- **AUROC**: P(sick scored above healthy); 0.5 = random; no threshold.
- **Expected cost**: weights error types (unit costs = 1−accuracy).

- **Calibration**: confidence 0.8 → right 80%. Bars **below** diagonal = **over-confident**. Discrimination ≠ calibration (AUROC 1 with all scores in 0.4–0.6). ECE = $\sum_m \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$.

Distance metrics

$$\text{HD} = \max\left(\max_{i\in\partial\Omega_S} \min_{j\in\partial\Omega_Y} d(i,j),\ \max_{i\in\partial\Omega_Y} \min_{j\in\partial\Omega_S} d(i,j) \right)$$

$$\text{ASSD} = \frac{\sum_{\partial S} \phi_{\partial Y} + \sum_{\partial Y} \phi_{\partial S}}{|\partial S| + |\partial Y|};\ \text{HD95} = \text{95th pct per}$$

direction, then max (**robust to outliers**). MASD = mean of the two directed means. In mm, lower is better. Tubular: clDice.

- **Steps**: one-hot → extract **boundary** → EDT of both **boundaries** → mask with the other **boundary** → reduce (max / P95 / mean). Upsampling is superfluous. Pass voxel spacing.
- **#EDT**: textbook $2 \times 2 \times (K{+}1)$ = **4(K+1)** (Dice 0; HD and ASSD 2 per class each, incl. background).
- **Faster**: skip background, share EDTs across HD/ASSD → **2K**; class missing → skip EDT, use default value. GPU: cuCIM, DisTorch.
- **Empty masks**: both empty → 0; one empty → undefined (default e.g. scan diagonal, biases results).
- $\partial\Omega_S = \Omega_S$: (1) **thin** object (line, single voxel, no interior); (2) **empty** mask, $\partial\emptyset = \emptyset$.
- **Shuffled slices**: Dice **unchanged**; ASSD/HD **broken** (random).
- 11 tools differ by >100 mm (boundary definition, empty masks, spacing): report tool and version.

6 ETHICS & REPRODUCIBILITY

Consent, purpose, care; anatomy itself can identify people; "reproducible" has several meanings.

- **GDPR**: consent, purpose, good care of data.
- **Retrospective use**: needs consent for that use; exceptions when it *cannot be asked* or *cannot reasonably be required* → **ask the ethics committee for an exemption** (deceased patients). NL: WMO, METC, internal board.
- **Data Management Plan** before collecting: security, quality, legal compliance.
- **Anonymisation vs pseudo-anonymisation**: both replace the ID by a random number; **pseudo keeps the mapping somewhere** (re-identifiable); anonymisation deletes it.
- **Ease of anonymisation**: biopsy < head MRI (deface) < whole-body CT (body shape) < **iris** (biometric, like a fingerprint).
- **Public challenges**: **common benchmark** (same test set and metrics) + **reproducibility** (public data and protocol).
- Licences: CC BY vs CC BY-NC (no commercial use). DTA for data outside your group. Code may be public, **patient data never**.

Reproducibility

- **4 meanings**: (1) bit-identical weights; (2) same code+data on another machine; (3) other package versions; (4) results within a confidence interval.
- **Randomness**: weight init, batch sampling, GPU non-determinism, hardware, kernel versions. Pseudo-random = deterministic given seed. "Testing a model, a seed, or an artifact?"
- **Fixes**: seed `random`, `numpy`, `torch`; `torch.use_deterministic_algorithms`; pin versions; report several seeds.
- **Public code, private data**: "reproducible" is ill-defined; **partially** reproducible; method reusable on other tasks.
- MIDL checklist: code, public data, dependencies, training and evaluation code, trained model, documentation, licence.
- Deployment: Therac-25; responsibility spread over data collectors, engineers, company, regulator, user; radiologist + AI > radiologist alone; Air France 447 (automation → lost intuition), Air Transat 236 (confirmation bias). Hygiene: collect less, report breach ASAP, 2FA, encrypted drives.

7 RECONSTRUCTION

Measure less (faster), let a model fill in the rest: plausible, not measured.

- **k-space**: centre = low frequencies (contrast, coarse shape); edges = high frequencies (detail). Under-sampling ×2 → overlapping (aliased) anatomy. Classical fix: **compressed sensing** (iterative).
- **Learned**: $N_\theta : \tilde{\mu} \mapsto \tilde{x}$, $\mathcal{L}_{rec} = \sum(\tilde{x} - x)^2$ against fully sampled images (simulated under-sampling).
- **VORTEX**: $\mathcal{L}_{rec} + \lambda\mathcal{L}_{cons}$; consistency between reconstructions of original and augmented input (no reference needed). k-space aug (noise, motion) → **invariant**; image aug (rotation) → **equivariant**.
- **Generative imaging**: denoising, super-resolution, modality translation, anomaly detection (pseudo-healthy reconstruction, $a = |x - \hat{x}_H|$). **Plausible ≠ measured.**

Four DL approaches

- **Conventional**: classical reconstruction, then DL on the image.
- **Direct**: one network, raw data → image, no physics.
- **Two-step / iterative**: unrolled; physics data-consistency + learned regulariser.
- **Raw-to-task**: raw data → segmentation or diagnosis, no image.
- **Choosing**: strong physics model + image needed → **iterative**; sparse data, little prior → **direct**; only a diagnosis needed, noisy data → **raw-to-task**.
- **Real time**: iterative slowest (fewer iterations or conventional); direct = one pass but memory heavy; raw-to-task fastest.

8 DOMAIN SHIFT, FEDERATED, FOUNDATION

Training population eq deployment population: adapt the model, or train without moving the data.

- **Domain shift**: train on D_1, test on D_2; from scanner, vendor, protocol, operator, population. Training longer does not help.
- **Transfer learning**: fine-tune on a few labelled target images (freeze early layers, small lr).
- **Domain adaptation** (unlabelled target): intensity harmonisation, feature alignment, **adversarial** domain classifier; augment contrast/noise. **Source-free**: only trained model + unlabelled target.
- **Federated learning**: server sends model → local training → send **weights only** → average (FedAvg, weighted by data size). Different scanners/populations → global model worse.
- **Differential privacy**: datasets differing by one person give indistinguishable outputs.
- **No free lunch**: FL and DP → lower performance, higher compute.
- **SAM**: image encoder + **prompt encoder** (points, boxes, text, masks) + mask decoder. 2D natural images; SAM2 video mode for 3D.
- **Fine-tuning trade-off**: in-domain ↑, generalisation ↓ → on private CT **SAM** > **MedSAM**.
- FM axes: promptable?, general/medical, fine-tuned?, supervised/self-supervised.
- Others: nnInteractive (point, box, lasso, scribble; stable under prompt variability), TotalSegmentator (CT + MRI), Virchow (pathology embeddings).

9 CLASSIFICATION & EXPLAINABILITY

The hard part is the question and the labels; huge images via MIL; prefer quantification over decisions.

- **Formulate the question**: binary, bins, or regression? Is the label reliable? ("everything is a maybe").
- **Expert panel** (Barrett, DEPP): >75% agree → consensus; experts disagree. **One net per rater** → virtual panel: interpretable, better calibrated (ECE 0.18 → 0.05), free uncertainty.
- **MIL**: positive bag = at least one positive instance. WSI → tiles → foundation-model embeddings → attention aggregator. "Smoking gun" (easy) vs spatial combination (hard).
- **Quantification over decision**: probability maps, measured displacement.
- **Many classes**: hierarchical (coarse → fine); reformulate (merge, bins, one-vs-rest, regression); embedding learning.
- **Shortcut learning**: model uses a confounder correlated with the label (implants, contrast, scanner). Fix: balance the confounder, mask to the organ, artefact augmentation, check saliency.
- **LIME**: random superpixel perturbations + local linear model. Unstable from random sampling; superpixels not anatomical; perturbed images unrealistic. Fix: more samples / average runs, anatomical segments, realistic perturbation, compare Grad-CAM/SHAP.
- **SHAP**: additive, prediction = baseline + $\sum$ feature contributions.

10 GUEST LECTURES

Priors (cardiac), data splits (drug discovery), product reality (orthopaedics), abstention and shortcuts (pathology).

Cardiac imaging (symmetry)

- Build what you know into the model. **Equivariant** $f(gx) = gf(x)$ (segmentation); **invariant** $f(gx) = f(x)$ (classification).
- CNNs are translation- but not rotation-equivariant; augmentation only approximates. Group conv adds an angle axis (x, y, θ). SE(2) = rotations + translations.
- CyclePhase: cardiac phase on a circle $(\cos\varphi, \sin\varphi)$. Neural fields $f(x, y)$: resolution chosen at query time.
- Equivariance breaks silently (pooling, normalisation): measure it.

Drug discovery (Atomwise)

- 3D CNN on voxelised protein + molecule. **Random split 99.9%** (near-duplicate molecules = leakage) → time split 90% → **target split** (by protein similarity) = real accuracy.
- Decoys for missing negatives; rotation augmentation favoured symmetric molecules; invariant GNN not much better.

Orthopaedics (AtMoves)

- "Dice 0.94, so what?": the model is 10% of the product.
- **Measure, don't classify**: CT under load → nnU-Net segmentation → registration → implant motion.
- Paper → prototype → software → certified → used → paid → better care. Intended use decides the regulatory class.
- Evidence: technical, clinical, **utility**. User ≠ payer ≠ saver. Patient-level splits.

Clonality (pathology)

- Two lung tumours: clonal (metastasis, palliative) vs independent (curative). NGS costs €1700/tumour, >1 week.
- Tiles → foundation model → **cross-attention** → attention MIL. **Precision first** (false "clonal" removes a cure).
- **Deep Gamblers**: extra abstain output, $\ell = -\log(p_y + p_{abs}/o)$, $1 < o < 2$ → ≈98% precision on the 20% answered.
- Shortcuts: scanner, size, subtype, **patient identity**.

COURSE THREADS

- Start from the clinical question: which decision does the output change?
- Data handling decides results: split by patient/centre/target; watch leakage and shortcuts.
- The metric must match the question; use several families; report uncertainty.
- Few labels → priors: symmetry, weak labels, foundation models.
- Corner cases (empty masks, 0/0) need explicit conventions; every convention biases.
- AI quantifies and supports; the clinician decides.

KEY NUMBERS

- nnU-Net: anisotropic if spacing ratio > 3; cascade if patch < 12.5%; batch ≥ 2; 1/3 fg patches; 5-fold CV; norm pct 0.5/99.5.
- U-Net 572 → 388; 4 poolings → size divisible by 16.
- HU: water 0, air −1000. Doses: chest CT 7 mSv, background 4 mSv/yr, 100 mSv/yr cancer risk.
- Inter-rater ≈90% DSC. DEPP consensus >75%. Virtual panel ECE 0.183 → 0.050.